

DSAA Lab 02

Sorting Techniques

Areeba (TCT-004)

Thar Institute of Engineering Sciences and Technology (TIEST)

CT-159: Data Structure Algorithms and Applications (Pr)

Engr. Aneel Kumar

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Exercise

Q1: Consider a list of students, their scores, and the date they took the test:

Name	Score	Date
Alice	85	2023-01-05
Bob	85	2023-01-03
Charlie	80	2023-01-04
David	90	2023-01-02

Sort this table by score.

Answer:

```
import java.util.*;

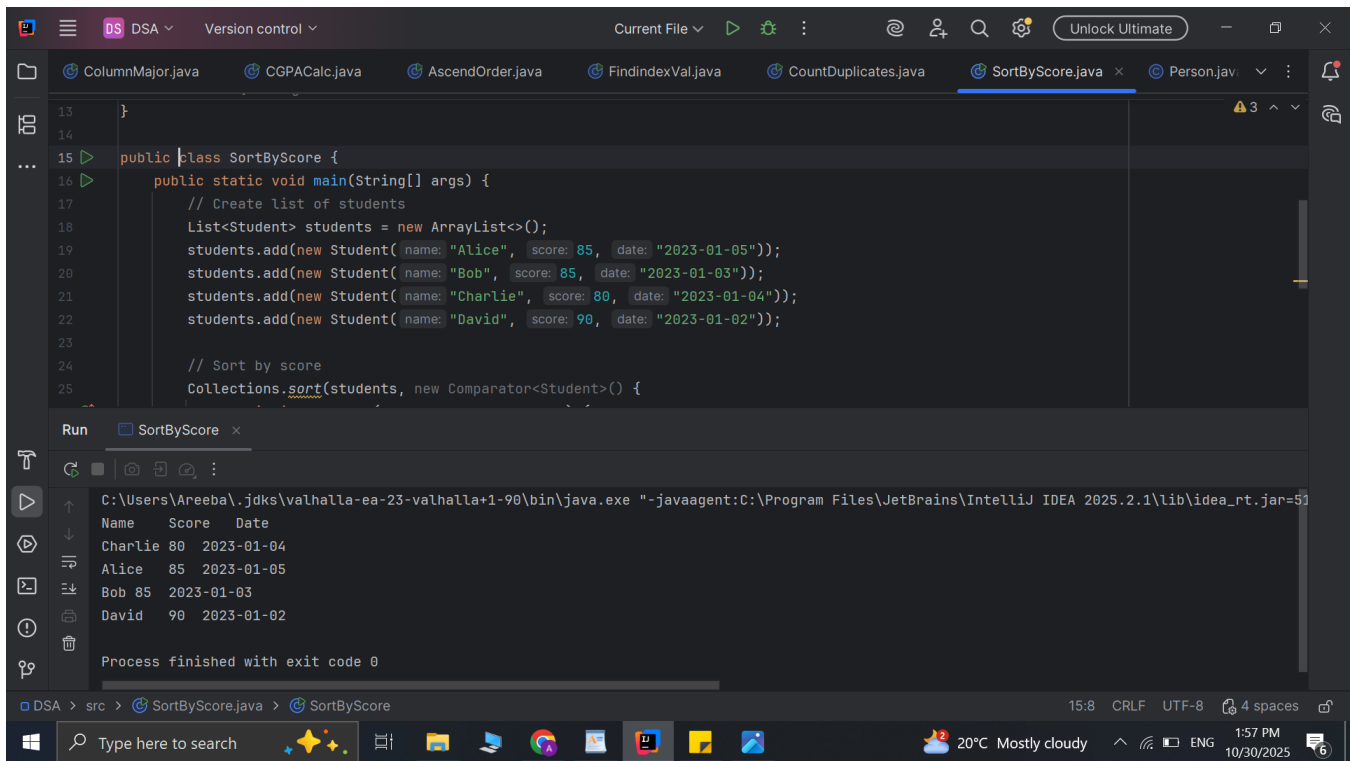
class Student {
    String name;
    int score;
    String date;

    Student(String name, int score, String date) {
        this.name = name;
        this.score = score;
        this.date = date;
    }
}

public class SortByScore {
    public static void main(String[] args) {
        // Create list of students
        List<Student> students = new ArrayList<>();
        students.add(new Student("Alice", 85, "2023-01-05"));
        students.add(new Student("Bob", 85, "2023-01-03"));
        students.add(new Student("Charlie", 80, "2023-01-04"));
        students.add(new Student("David", 90, "2023-01-02"));

        // Sort by score
        Collections.sort(students, new Comparator<Student>() {
            public int compare(Student s1, Student s2) {
                return Integer.compare(s1.score, s2.score);
            }
        });

        // Display sorted list
        System.out.println("Name\tScore\tDate");
        for (Student s : students) {
            System.out.println(s.name + "\t" + s.score + "\t" + s.date);
        }
    }
}
```



```
13 }
14
15 public class SortByScore {
16     public static void main(String[] args) {
17         // Create list of students
18         List<Student> students = new ArrayList<>();
19         students.add(new Student( name: "Alice", score: 85, date: "2023-01-05"));
20         students.add(new Student( name: "Bob", score: 85, date: "2023-01-03"));
21         students.add(new Student( name: "Charlie", score: 80, date: "2023-01-04"));
22         students.add(new Student( name: "David", score: 90, date: "2023-01-02"));
23
24         // Sort by score
25         Collections.sort(students, new Comparator<Student>() {
```

Run SortByScore

C:\Users\Areeba\.jdk\valhalla-ea-23-valhalla+1-90\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.2.1\lib\idea_rt.jar=51

Name	Score	Date
Charlie	80	2023-01-04
Alice	85	2023-01-05
Bob	85	2023-01-03
David	90	2023-01-02

Process finished with exit code 0

DSA > src > SortByScore.java > SortByScore 15:8 CRLF UTF-8 4 spaces 1:57 PM 10/30/2025

Q2: Create a Person class which has following attributes:

First Name
Last Name
Birth Year
Birth Month
Birth Date

Develop solution such that day month and year are taken as input for N persons and perform Sorting based on year, month and day using Selection Sort.

Answer:

```
import java.util.Scanner;

class Person {
    String firstName, lastName;
    int year, month, day;
}

public class SortPersons {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of persons: ");
        int n = sc.nextInt();

        Person[] p = new Person[n];

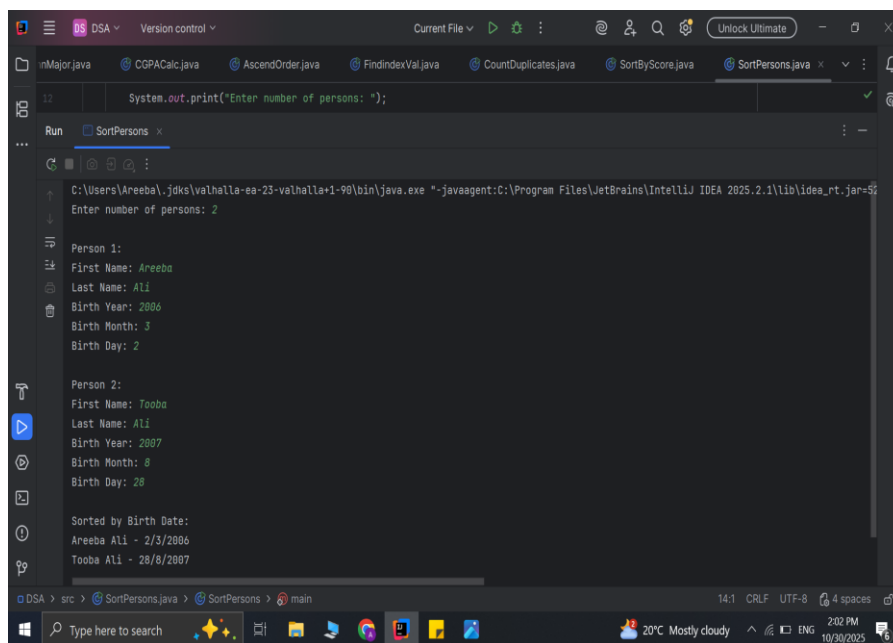
        // Input
        for (int i = 0; i < n; i++) {
            p[i] = new Person();
            System.out.println("\nPerson " + (i + 1) + ":");
            System.out.print("First Name: ");
            p[i].firstName = sc.next();
            System.out.print("Last Name: ");
            p[i].lastName = sc.next();
            System.out.print("Birth Year: ");
            p[i].year = sc.nextInt();
            System.out.print("Birth Month: ");
            p[i].month = sc.nextInt();
            System.out.print("Birth Day: ");
            p[i].day = sc.nextInt();
        }

        // Selection Sort by year, then month, then day
        for (int i = 0; i < n - 1; i++) {
            int min = i;
            for (int j = i + 1; j < n; j++) {
                if (p[j].year < p[min].year ||
                    (p[j].year == p[min].year && p[j].month < p[min].month) ||
                    (p[j].year == p[min].year && p[j].month == p[min].month && p[j].day <
p[min].day)) {
                    min = j;
                }
            }
            // swap

```

```
        Person temp = p[i];
        p[i] = p[min];
        p[min] = temp;
    }

    // Output
    System.out.println("\nSorted by Birth Date:");
    for (Person per : p) {
        System.out.println(per.firstName + " " + per.lastName + " - " +
            per.day + "/" + per.month + "/" + per.year);
    }
}
```



The screenshot displays the IntelliJ IDEA IDE with the file `SortPersons.java` open. The Run console shows the following output:

```
C:\Users\Areeba\jdk\valhalla-ea-23-valhalla-1-99\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.2.1\lib\idea_rt.jar=52
Enter number of persons: 2

Person 1:
First Name: Areeba
Last Name: Ali
Birth Year: 2006
Birth Month: 3
Birth Day: 2

Person 2:
First Name: Tooba
Last Name: Ali
Birth Year: 2007
Birth Month: 8
Birth Day: 28

Sorted by Birth Date:
Areeba Ali - 2/3/2006
Tooba Ali - 28/8/2007
```

The status bar at the bottom indicates the file is `SortPersons.java` in the `main` package, with a line number of 14:1, CRLF line endings, UTF-8 encoding, and 4 spaces for indentation.

Q3: Given an Integer n and a list arr. Sort the array using bubble sort algorithm.

Examples:

Input: n = 5, arr [] = {4, 1, 3, 9, 7}

Output: 1 3 4 7 9

Input: n = 10, arr [] = {10, 9, 8, 7, 6, 5, 4, 3, 2, 1}

Output: 1 2 3 4 5 6 7 8 9 10

Answer:

```
class BubbleSort {
    // Function to perform Bubble Sort
    public static void bubbleSort(int arr[], int n) {
        for (int i = 0; i < n - 1; i++) {
            for (int j = 0; j < n - i - 1; j++) {
                if (arr[j] > arr[j + 1]) {
                    // Swap arr[j] and arr[j+1]
                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                }
            }
        }
    }

    // Main function just to test the code
    public static void main(String[] args) {
        int arr[] = {4, 1, 3, 9, 7};
        int n = arr.length;

        bubbleSort(arr, n);

        // Print sorted array
        for (int num : arr)
            System.out.print(num + " ");
    }
}
```

